PORTFOLIO

Benihckey Viara Enysya

Data Analyst &
Business Intelligence

Hi! I'm **BENIHCKEY VIARA ENYSYA**



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MBA graduate at Bandung Institute of Technology (ITB) with a background in Architecture at BINUS University and Asset Management experience at Sinar Mas Land. Building hands-on data analytics and business intelligence skills, including Excel, SQL/Python, data cleaning, exploratory data analysis, and dashboard development, through a dedicated bootcamp, complemented by MBA coursework in supply chain, operations, and data-driven decision-making. Seeking opportunities in data analysis, business intelligence, and business analytics.

EDUCATION

2026 - PRESENT	**DATA ANALYTICS & BUSINESS INTELLIGENCE, DIBIMBING**
2025 - 2026	**BANDUNG INSTITUTE OF TECHNOLOGY, MBA** *GPA: 3.74/4.00 (49 CREDITS)*
2020 - 2024	**BINUS UNIVERSITY, BACHELOR DEGREE IN ARCHITECTURE** *GPA: 3.54/4.00 (146 CREDITS)*
2017 - 2020	**SMAN 2 TANGERANG, HIGH SCHOOL DIPLOMA IN SCIENCE**

WORK EXPERIENCE

OCT 2024 - MAR 2025	**FIT-OUT INTERN IN ASSET MANAGEMENT, SINAR MAS LAND** *SOUTH TANGERANG, BANTEN*
FEB 2023 - JUN 2023	**ARCHITECTURAL PLANNING INTERN, SINAR MAS LAND** *SOUTH TANGERANG, BANTEN*
FEB 2022 - FEB 2023	**EDUCATION MENTOR, BINUS UNIVERSITY** *WEST JAKARTA, SPECIAL CAPITAL REGION OF JAKARTA*

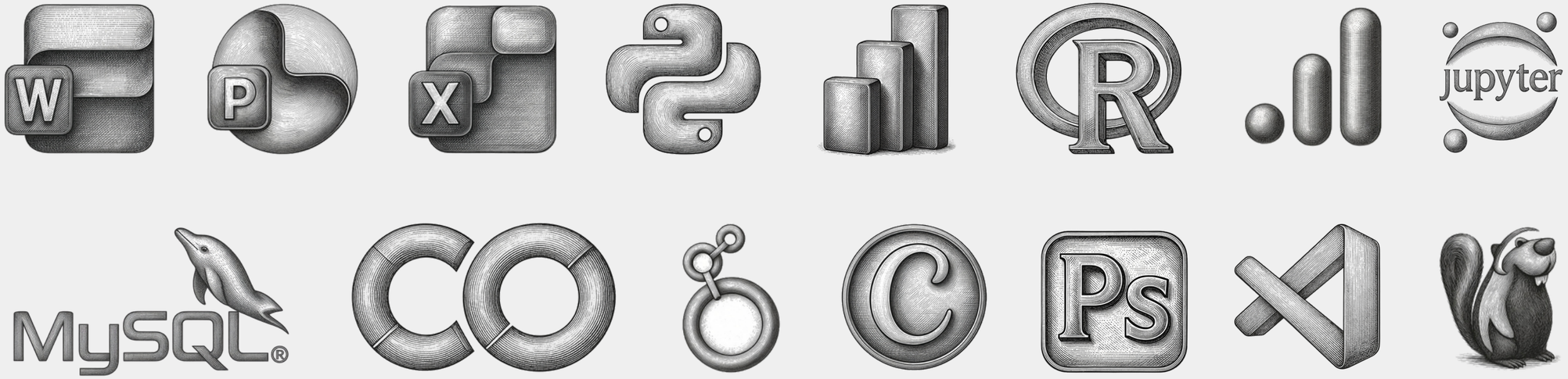
ORGANIZATION EXPERIENCE

SEP 2023 - JAN 2024	**UNDERGRADUATE RESEARCHER, UNIVERSITY OF INDONESIA**
JUL 2022 - AUG 2022	**TREASURER OF TFISC MERCHANDISE PROGRAM, TFISC**
MAY 2022 - OCT 2022	**CREATIVE DESIGN STAFF OF LITERATURE DAY 2022, TFISC**
MAY 2022 - AUG 2022	**TREASURER OF GREENER DAY 2022, TFISC**
APR 2022 - NOV 2022	**EXHIBITION RESOURCE TEAM COORDINATOR OF INTERSEMINARCH 2022, HIMARS BINUS**
MAR 2022 - MAR 2023	**CREATIVE DESIGN ACTIVIST, TFISC**
MAY 2021 - DEC 2021	**EXHIBITION RESOURCE TEAM STAFF OF ARCHITECTURE GRAND FESTIVAL 2021, HIMARS BINUS**

SOFT SKILLS

- Cross-Department Communication
- Active Listening
- Stakeholder Reporting
- Technical Writing
- Team Collaboration
- Mentoring & Coaching
- Time Management
- Multitasking
- Project Coordination
- Budget Management
- Analytical Thinking
- Attention to Detail
- Adaptability
- Emotional Intelligence
- Continuous Learning
- Resilience Under Pressure

HARD SKILLS



LANGUAGE

INDONESIAN
ENGLISH



DATA ANALYST & BUSINESS INTELLIGENCE **PROJECT**

E-Commerce Sales Performance & Customer Segmentation (**RFM Analysis**)

GitHub Link

DATASET OVERVIEW

Dataset - Kaggle ‘Online Retail Dataset‘

index	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	554697	21166	COOK WITH WINE METAL SIGN	1	5/25/2011 17:31	2.08	14584	United Kingdom
1	561038	82482	WOODEN PICTURE FRAME WHITE FINISH	2	7/24/2011 11:58	2.55	17114	United Kingdom
2	560552	23192	BUNDLE OF 3 ALPHABET EXERCISE BOOKS	1	7/19/2011 12:54	1.65	15311	United Kingdom
3	559884	85183B	CHARLIE & LOLA WASTEPAPER BIN FLORA	12	7/13/2011 11:34	1.25	16843	United Kingdom
4	544450	21789	KIDS RAIN MAC PINK	3	2/20/2011 12:23	0.85	17811	United Kingdom

TOTAL TRANSACTIONS

4,870

Across 12 months

UNIQUE CUSTOMERS

1,932

Analyzed for segmentation

TIME PERIOD

Dec 2010 - Dec 2011

1 Full Year Snapshot

Data Dictionary & Types

- InvoiceNo [Int]: 6-digit transaction ID.
- StockCode [Str]: 5-digit product item code.
- Description [Str]: Product item name.
- Quantity [Int]: Item units per transaction.
- InvoiceDate [Datetime]: Transaction timestamp.
- UnitPrice [Float]: Product price per unit.
- CustomerID [Int]: 5-digit customer ID.
- Country [Str]: Customer's country of residence.

EXECUTIVE RECOMMENDATIONS

Strategic Action Plan based on RFM Segmentation & Sales Trends



RETENTION

VIP / Champion Segment

Prioritize a tiered **loyalty program for the VIP segment**. Provide early product access and exclusive rewards to maintain their exceptionally high Customer Lifetime Value (CLV).



REACTIVATION

At-Risk / Hibernating

Launch targeted **email marketing campaigns** with aggressive, time-sensitive discounts to re-engage the high number of customers who have not transacted recently.



OPERATIONAL FOCUS

United Kingdom Market

Given the **UK market's massive dominance** (>80% of revenue), optimize supply chain and fulfillment center logistics specifically for the UK region to reduce overhead costs.

PROBLEM STATEMENT

Unlocking Growth Through Data-Driven Customer Personalization

BUSINESS CONTEXT

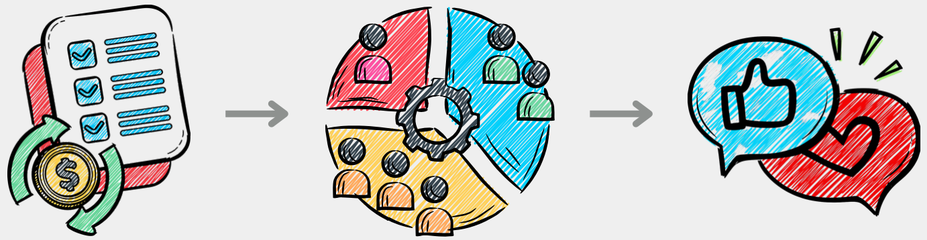
The company experienced irregular revenue fluctuations over the 12-month period. Traditional mass marketing is no longer cost-effective, demanding a swift transition to personalized, data-driven customer strategies.

RESEARCH QUESTION

How can we classify our 1,932 active customers into distinct, behavior-based segments based on their transactional history to enable highly targeted, high-return marketing campaigns?

THE SOLUTION STRATEGY

Transitioning to Data-Driven Marketing



Raw Transaction Data → **Data-Driven Segmentation** → **Tailored Engagement**

GOAL

Convert 1,932 diverse customers into specific actionable groups to maximize ROI.

DATA PREPROCESSING

Ensuring Data Integrity and Engineering Features for RFM Analysis

01. DATA CLEANSING

The dataset is exceptionally clean, requiring minimal intervention. All records are valid for segmentation.

- Zero Missing Values (Nulls)
- Zero Duplicated Rows
- Zero Anomalies (No negative quantities)
- 4,870 Valid Transactions Maintained

02. FEATURE ENGINEERING

Transforming raw data columns into actionable metrics required for the Time-Series and RFM clustering models.

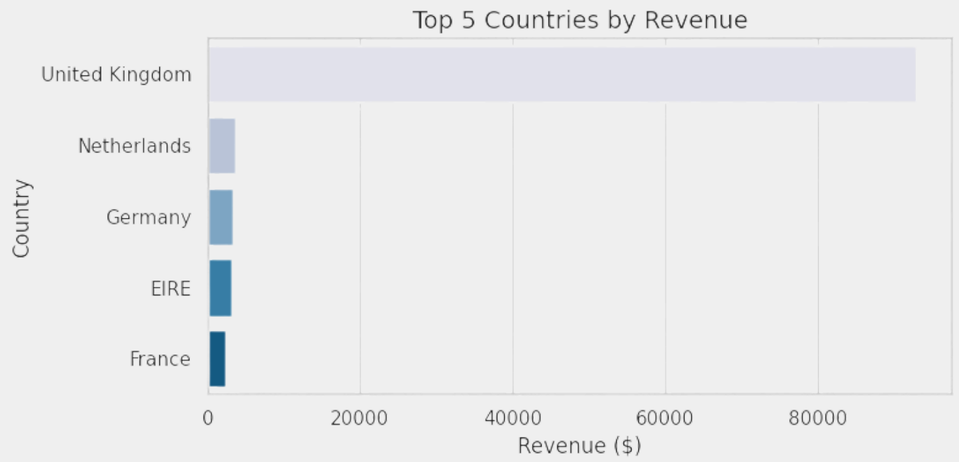
Monetary Metric Creation
TotalSales = Quantity × UnitPrice

Time-Series Preparation
InvoiceDate (String) -> Datetime Format

EXPLORATORY DATA ANALYSIS

Discovering Market Dominance and Seasonal Purchasing Trends

MARKET DOMINANCE



The **UK heavily dominates sales**, generating >80% of total revenue.

Action: *Optimize UK fulfillment logistics for cost-efficiency, while actively diversifying into secondary EU markets to reduce geographic concentration risk.*

SEASONALITY & TRENDS



Sales peak sharply in November, driven by early holiday gift purchasing.

Action: *Front-load marketing budgets and inventory stockpiling in October and November to capture this peak demand window.*

RFM ANALYSIS FRAMEWORK

Defining the Core Behavioral Metrics for Customer Segmentation

RFM is a behavioral model used to analyze customer value by measuring when they bought (Recency), how often they buy (Frequency), and how much they spend (Monetary). It is a highly cost-effective method to drive targeted marketing.

Recency

How many days since the customer's last purchase?

A lower recency indicates an active, engaged customer who is more likely to respond to new promotions.

Frequency

How many times has the customer purchased in total?

A higher frequency indicates loyalty and satisfaction with the company's product offerings.

Monetary

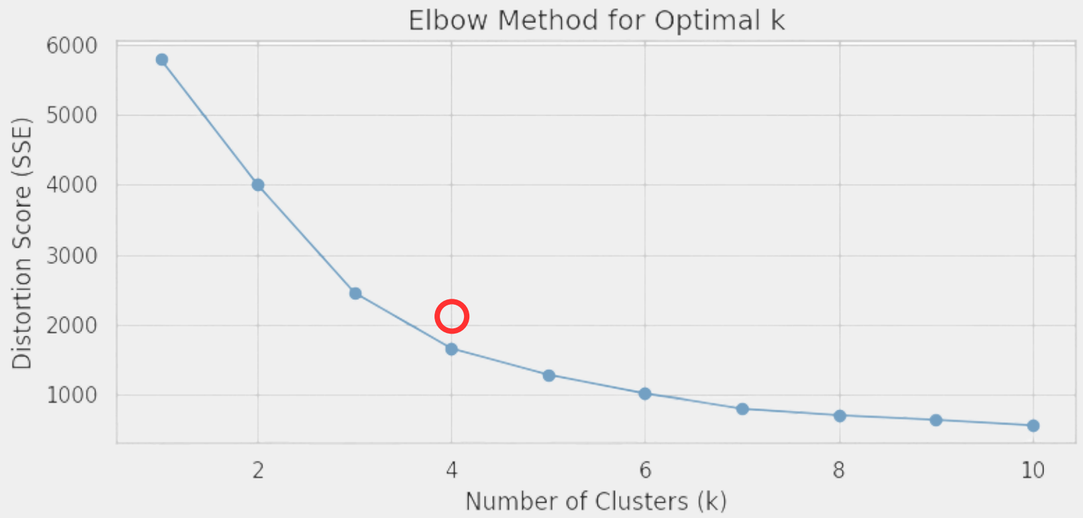
How much total revenue has the customer generated?

A higher monetary value identifies the most profitable customers who drive the business's bottom line.

MODEL EVALUATION

Determining the Optimal Customer Segments with K-Means Clustering

The Elbow Method



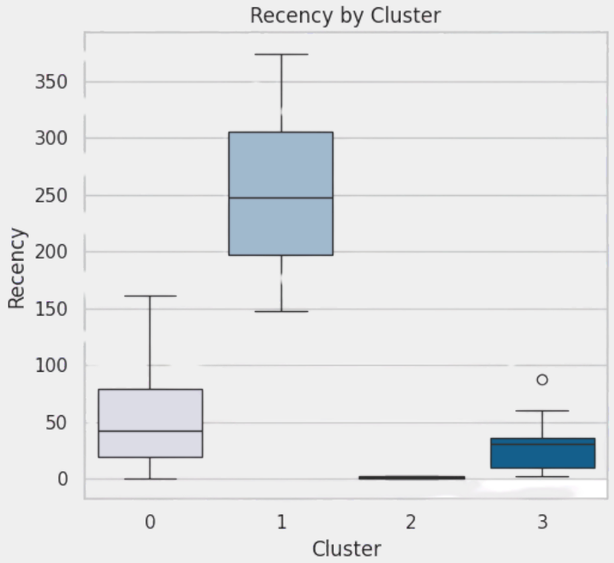
01. STANDARDIZATION
RFM metrics were standardized (scaled) to ensure equal weight before feeding them into the algorithm.

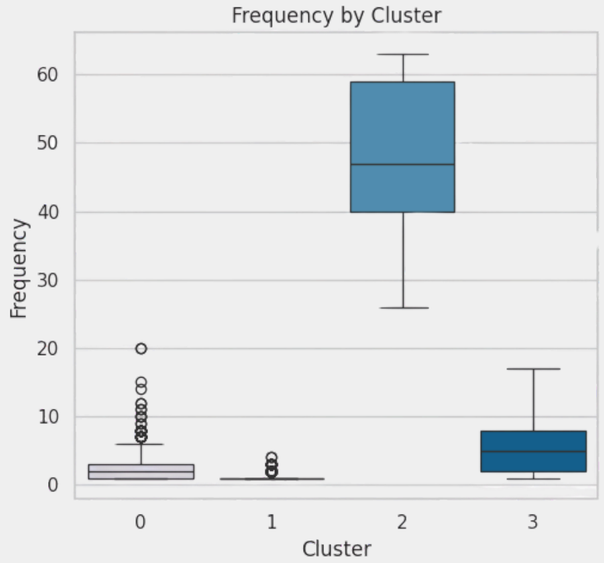
02. CLUSTERING ALGORITHM
The K-Means machine learning algorithm was applied to group customers with statistically similar purchasing behaviors.

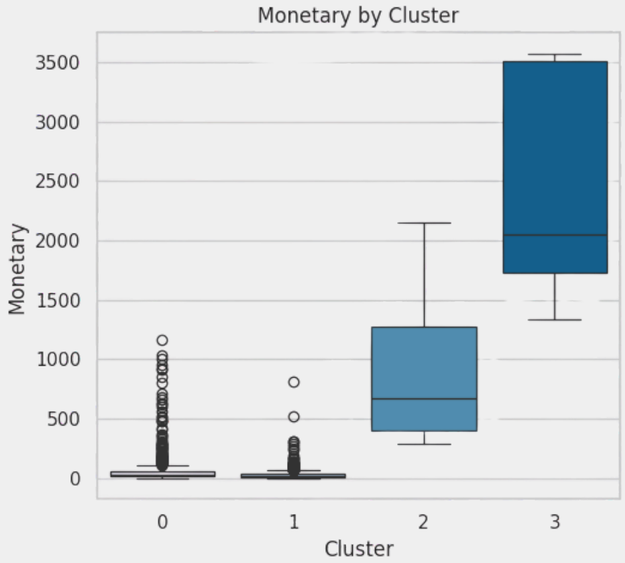
03. OPTIMAL DECISION (K=4)
The curve 'elbows' at K=4. This provides a distinct, actionable categorization of the customer base without overcomplicating the marketing strategy.

INSIGHT & ANALYSIS

Translating K-Means Clusters into Actionable Business Profiles







PROMISINGS
(Cluster 0)
Steady Frequency

The largest customer base (1,258). They purchase steadily with moderate recency.

AT-RISK / HIBERNATING
(Cluster 1)
High Recency Lapse

Have not transacted in a long time (Avg 252 days) and have the lowest overall spend.

FREQUENT BUYERS
(Cluster 2)
Extreme Volume

Exceptional purchase frequency (Avg 47 orders) and very recent activity. Highly engaged.

VIP CHAMPIONS
(Cluster 3)
Highest Monetary Value

Spent heavily (Avg $2,446) and purchased recently. They are the primary drivers of revenue.

THANK YOU!


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